Dominance of the shifted weight along minimal-length coset representatives

Lecouvey, Cédric, Wahiche, David

doi:10.3842/SIGMA.2025.023; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2025-023-macdonald-identities-weyl-kac-denominator-fo/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author Lemma environment and its own proof environment. The span contains no citation and no reference, so no stub and no bibliography is retained; the author's own qed closing (qedhere inside the final display, tag*{\qed}, \renewcommand{\qed}{}) is kept verbatim.

Editorial note. Printed numbers verified against the source declarations and a full-file walk: the Theorem family counter is per-section (family shared) and the equation counter is per-section (numberwithin); this lemma is the fourth family environment of section 4 (after 4.1 at line 681, 4.2 at 728, 4.3 at 773) and one numbered display (source line 753) precedes the span in section 4, so initial_counters section 4, Theorem 3, equation 1 reproduce the printed number 4.4. All displays inside the span are unnumbered, so there is no in-span drift.

Lemma 4.4. *For any dominant weight $\lambda \in M^*$ and any $u \in W^\lambda$ (i.e., of minimal length in the cosets of W/W_λ), the weight*

$$\eta^\vee\lambda + \big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big)$$

is dominant.

Proof. We need to prove that

$$\big(\eta^\vee\lambda + \big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i\big) = \big(\eta^\vee(\lambda, \alpha_i) + \big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i\big) \geq 0$$

for any $i \in I^*$. Observe that we have

$$\big(\big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i\big) = \big(u^{-1}(\mathring{\rho}), \alpha_i\big) - (\mathring{\rho}, \alpha_i) = (\mathring{\rho}, u(\alpha_i)) - \frac{1}{2}\|\alpha_i\|^2$$

since $(\mathring{\rho}, \alpha_i) = \frac{\|\alpha_i\|^2}{2}\big(\mathring{\rho}, \alpha_i^\vee\big) = \frac{\|\alpha_i\|^2}{2}$. If $u(\alpha_i) \in R_+$, then $(\mathring{\rho}, u(\alpha_i)) \geq \frac{1}{2}\|\alpha_i\|^2$ and therefore $\big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i) \geq 0$. We are done because λ is dominant and thus $\eta^\vee(\lambda, \alpha_i) \geq 0$. Now assume $u(\alpha_i) \in -R_+$ is a negative root. Then, we know that there exists a reduced expression of u ending by s_i, that is of the form $u = u's_i$ with $\ell(u) = \ell(u') + 1$. Since $u \in W^\lambda$, this implies that $(\lambda, \alpha_i) \geq p$ where $p = 1$ in all affine types except in types $C_n^{(1)}$ and $G_2^{(1)}$ where $p = 2$ and

$p = 3$, respectively (because $\lambda \in M^* \cap P_+$). Indeed, we would have otherwise $u(\lambda) = u' s_i(\lambda) = u'$ and u would not be of minimal of length. We thus have

$$\big(\eta^\vee \lambda + \big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i\big) \geq p\eta^\vee - (\mathring{\rho}, -u(\alpha_i)) - \frac{1}{2}\|\alpha_i\|^2 \geq 0,$$

where $\alpha \in R_+$. One can then check that for any $\alpha \in R_+$, we have $(\mathring{\rho}, -u(\alpha_i)) + \frac{1}{2}\|\alpha_i\|^2 \leq p\eta^\vee$ and thus the desired inequality

$$\big(\eta^\vee \lambda + \big(u^{-1}(\mathring{\rho}) - \mathring{\rho}\big), \alpha_i\big) \geq 0. \qquad \square \quad \square$$